

❤️ Heart Disease Prediction – End-to-End MLOps Pipeline

📌 Overview

This project implements an **end-to-end MLOps pipeline** to predict the presence of heart disease using the **UCI Heart Disease dataset**.

The solution demonstrates the complete lifecycle of a production-grade ML system including preprocessing, training, experiment tracking, model versioning, and API-based inference.

🧠 Problem Statement

Predict whether a patient has heart disease based on clinical and diagnostic attributes using a binary classification model.

📊 Dataset

- **Name:** UCI Heart Disease Dataset (Cleveland subset)
 - **Records:** 303 (after cleaning: 297)
 - **Features:** 14
 - **Target:** Heart disease (0 = No, 1 = Yes)
-

📁 Project Structure

heart-disease-mlops/

├ data/

│ └ raw/raw_heart_data.csv

│ └ processed/heart.csv

├ src/

│ └ preprocess.py

│ └ train.py

│ └ app.py

├ notebooks /

│ └ 00_clean_data.ipynb

- └ 01_eda.ipynb
- └ 02_training.ipynb
- └ 03_inference.ipynb
- └ tests/
 - └ test_preprocess.py
 - └ test_train.py
- └ model/
 - └ development/
 - └ {version:v#}/
 - └ heart_model.pkl
 - └ production/
 - └ {version:v#}/
 - └ heart_model.pkl
- └ mlruns/
- └ requirements.txt
- └ pytest.ini
- └ README.md

Setup Instructions

```
python -m venv venv
```

```
venv\Scripts\activate
```

```
pip install -r requirements.txt
```

Testing

Run all unit tests:

```
pytest
```

Model Training

Train the model and log experiments using MLflow:

```
python -m src.train
```

MLflow Experiment Tracking

Start MLflow UI:

```
mlflow ui
```

Access at: <http://localhost:5000>

Model Versioning

Each training run creates a new model version:

```
model/{env}/
```

```
└─ v1/
```

```
└─ v2/
```

FastAPI Inference Service

Start the API:

```
uvicorn src.app:app --reload
```

Swagger UI: <http://127.0.0.1:8000/docs>

API Contract

Endpoint

POST /predict

Request

```
{  
  "age": 63,  
  "sex": 1,
```

```
"cp": 1,  
"trestbps": 145,  
"chol": 233,  
"fbs": 1,  
"restecg": 2,  
"thalach": 150,  
"exang": 0,  
"oldpeak": 2.3,  
"slope": 3,  
"ca": 0,  
"thal": 6  
}
```

Response

```
{  
  "heart_disease": 0,  
  "confidence": 0.3271  
}
```



Key MLOps Concepts

- Reproducible pipelines
- Unit testing
- Experiment tracking
- Model versioning
- Training–serving consistency
- API-based inference



Conclusion

This project demonstrates a complete MLOps workflow from raw data to a deployable ML service, emphasizing reproducibility, reliability, and production readiness.